[INCH-POUND]
A-A-59416
18 February 2004

COMMERCIAL ITEM DESCRIPTION

INDUCTOR, CHIP, FIXED, HIGH SELF RESONANT FREQUENCY, SURFACE MOUNT

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers the general requirements for a radio frequency coil. Coils covered by this CID are intended for commercial/industrial applications.

2. CLASSIFICATION. This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see 7.1).

AA59416	-	001		K
CID number		Dash number (see table I)		Inductance Tolerance (see table II)

3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. Inductors supplied to this CID shall be as specified herein. (see figure 1).

3.2. Electrical characteristics. The electrical characteristics shall be as specified in table I.

3.3 Weight. The weight shall be no greater than 0.5 gram maximum.

3.4 Operating temperature range. The operating temperature range is –40°C to +125°C.

3.5 Temperature rise. DC current rating for a 15°C rise.

3.6 Altitude. The maximum altitude is 70,000 feet.

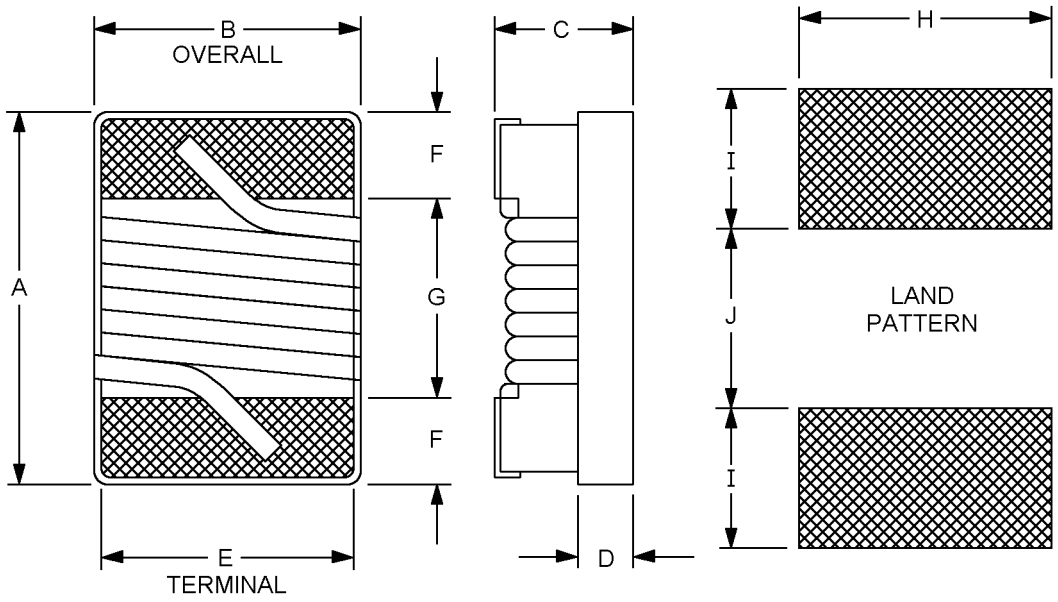
3.7 Marking. Inductors supplied to this CID shall be marked with the manufacturer's standard commercial PIN.

Beneficial comments recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAT, Post Office Box 3990, Columbus, OH 43216-5000, or telephone (614) 692-0557, or facsimile (FAX) (614) 692-6939.

AMSC N/A FSC 5950

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Dimension	Inches	mm
A Max.	.071	1.80
B Max.	.044	1.12
C Max.	.040	1.02
D Ref.	.015	0.38
E	.030	0.76
F	.013	0.33
G	.034	0.86
H	.040	1.02
I	.025	0.64
J	.025	0.64

NOTES:

1. Dimensions are in Inches.
2. Metric equivalents are listed for general information only.

FIGURE 1. Configuration and dimensions.

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TABLE I. Electrical characteristics.

CID dash number AA59416-	Inductance(nH)	Tolerance (±%)	Q Min	900 MHz		1.7 GHz		SRF Min (MHz)	DCR Max (OHMS)	1/ I_{DC}MAX (mA)
				L Typ	Q Typ	L Typ	Q Typ			
001	1.6 @ 250 MHz	10, 5	24	1.67	49	1.65	63	12,500	0.030	700
002	1.8 @ 250 MHz	10, 5	16	1.63	35	1.66	50	12,500	0.045	700
003	3.3 @ 250 MHz	10, 5, 2	35	3.31	75	3.38	88	5900	0.045	700
004	3.6 @ 250 MHz	10, 5, 2	22	3.72	53	3.71	65	5900	0.063	700
005	3.9 @ 250 MHz	10, 5, 2	22	3.95	49	3.96	67	6900	0.080	700
006	4.3 @ 250 MHz	10, 5, 2	22	4.32	50	4.33	70	5900	0.063	700
007	4.7 @ 250 MHz	10, 5, 2	20	4.72	47	4.75	57	5800	0.116	700
008	5.1 @ 250 MHz	10, 5, 2	20	4.93	47	4.95	56	5700	0.140	700
009	5.6 @ 250 MHz	10, 5, 2	26	5.77	63	6.05	80	4760	0.075	700
010	6.8 @ 250 MHz	10, 5, 2	27	6.75	60	7.10	81	5800	0.110	700
011	7.5 @ 250 MHz	10, 5, 2	28	7.70	60	7.82	65	4800	0.106	700
012	8.2 @ 250 MHz	10, 5	30	8.25	82	8.37	87	4200	0.115	700
013	8.7 @ 250 MHz	5, 2	28	8.86	62	9.32	58	4600	0.109	700
014	9.5 @ 250 MHz	5, 2	28	9.7	59	9.92	61	5400	0.135	700
015	10 @ 250 MHz	5, 2	31	10.0	66	10.6	83	4800	0.130	700
016	11 @ 250 MHz	5, 2	30	11.0	53	11.5	56	4000	0.086	700
017	12 @ 250 MHz	5, 2	35	12.3	72	13.5	83	4000	0.130	700
018	15 @ 250 MHz	5, 2	35	15.4	64	16.8	89	4000	0.170	700
019	16 @ 250 MHz	5, 2	34	6.2	55	17.3	52	3300	0.104	700
020	18 @ 250 MHz	5, 2	35	18.7	70	21.4	69	3100	0.170	700
021	22 @ 250 MHz	5, 2	38	22.8	73	26.1	71	3000	0.190	700
022	23 @ 250 MHz	5, 2	38	24.1	71	28.0	67	2850	0.190	700
023	24 @ 250 MHz	5, 2	36	24.5	45	28.7	39	2650	0.135	700
024	27 @ 250 MHz	5, 2	40	29.2	74	34.6	65	2800	0.220	600
025	30 @ 250 MHz	5, 2	37	31.4	47	39.9	28	2250	0.144	600
026	33 @ 250 MHz	5, 2	40	36.0	67	49.5	42	2300	0.220	600
027	36 @ 250 MHz	5, 2	37	39.4	47	52.7	24	2080	0.250	600
028	39 @ 250 MHz	5, 2	40	42.7	60	60.2	40	2200	0.250	600
029	43 @ 250 MHz	5, 2	38	47.0	44	64.9	21	2000	0.280	600
030	47 @ 200 MHz	5, 2	38	52.2	62	77.2	35	2000	0.280	600
031	51 @ 200 MHz	5, 2	35	55.5	69	82.2	34	1900	0.270	600
032	56 @ 200 MHz	5, 2	38	62.5	56	97.0	26	1900	0.310	600
033	68 @ 200 MHz	5, 2	37	80.5	54	168	21	1700	0.340	600
034	72 @ 150 MHz	5, 2	34	82.0	53	135	20	1700	0.490	400

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TABLE I. Electrical characteristics – continued.

CID dash number AA59416-	Inductance(nH)	Tolerance (±%)	Q Min	900 MHz		1.7 GHz		SRF Min (MHz)	DCR Max (OHMS)	1/ I_{DC}MAX (mA)
				L Typ	Q Typ	L Typ	Q Typ			
035	82 @ 150 MHz	5, 2	34	96.2	54	177	21	1700	0.540	400
036	100 @ 150 MHz	5, 2	34	124	49	--	--	1400	0.580	400
037	110 @ 150 MHz	5, 2	32	138	43	--	--	1350	0.610	300
038	120 @ 150 MHz	5, 2	32	166	39	--	--	1300	0.650	300
039	150 @ 150 MHz	5, 2	28	250	25	--	--	990	0.920	280
040	180 @ 100 MHz	5, 2	25	305	22	--	--	990	1.25	240
041	200 @ 100 MHz	10, 5	25	--	--	--	--	900	1.98	200
042	210 @ 100 MHz	10, 5, 2	27	--	--	--	--	895	2.06	200
043	220 @ 100 MHz	5, 2	25	--	--	--	--	900	2.10	200
044	250 @ 100 MHz	10, 5, 2	25	--	--	--	--	822	3.55	120
045	270 @ 100 MHz	5, 2	24	--	--	--	--	900	2.30	170
046	330 @ 100 MHz	5, 2	25	--	--	--	--	900	3.89	100
047	390 @ 100 MHz	5	25	--	--	--	--	900	4.35	100

1/ DC current rating for a 15°C rise.

TABLE II. Inductance tolerances

Code	Tolerance (percentage)
K	±2
M	±5
N	±10

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

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5.2 Market acceptance. The following market acceptance criteria are necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years.

b. The company must have sold 1,000 units meeting this CID in the commercial marketplace over the past 2 years.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format example.

7.2 Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these inductors to DSCC under the Military Parts Control Advisory Group (MPCAG) evaluation program, CAGE code 58536 should be used.

7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

c. Packaging requirements.

7.4 Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
02113	Coilcraft Inc 1102 Silver Lake Road Cary, Illinois 60013-1658 Phone number (847) 639-2361 Uniform Resource Locator (URL): www.coilcraft.com

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7.5 Part number (P/N) supersession data. This CID supersedes the following manufacturers' P/N's as shown. This information is being provided to assist in reducing proliferation in the government inventory system.

TABLE III. P/N supersession data. 1/

CID dash number (see table I) AA59416-	Vendor commercial PIN 2/ CAGE 02113	CID dash number (see table I) AA59416-	Vendor commercial PIN 2/ CAGE 02113
001	0603CS-1N6X*B*	025	0603CS-30NX*B*
002	0603CS-1N8X*B*	026	0603CS-33NX*B*
003	0603CS-3N3X*B*	027	0603CS-36NX*B*
004	0603CS-3N6X*B*	028	0603CS-39NX*B*
005	0603CS-3N9X*B*	029	0603CS-43NX*B*
006	0603CS-4N3X*B*	030	0603CS-47NX*B*
007	0603CS-4N7X*B*	031	0603CS-51NX*B*
008	0603CS-5N1X*B*	032	0603CS-56NX*B*
009	0603CS-5N6X*B*	033	0603CS-68NX*B*
010	0603CS-6N8X*B*	034	0603CS-72NX*B*
011	0603CS-7N5X*B*	035	0603CS-82NX*B*
012	0603CS-8N2X*B*	036	0603CS-R10X*B*
013	0603CS-8N7X*B*	037	0603CS-R11X*B*
014	0603CS-9N5X*B*	038	0603CS-R12X*B*
015	0603CS-10NX*B*	039	0603CS-R15X*B*
016	0603CS-11NX*B*	040	0603CS-R18X*B*
017	0603CS-12NX*B*	041	0603CS-R20X*B*
018	0603CS-15NX*B*	042	0603CS-R21X*B*
019	0603CS-16NX*B*	043	0603CS-R22X*B*
020	0603CS-18NX*B*	044	0603CS-R25X*B*
021	0603CS-22NX*B*	045	0603CS-R27X*B*
022	0603CS-23NX*B*	046	0603CS-R33X*B*
023	0603CS-24NX*B*	047	0603CS-R39X*B*
024	0603CS-27NX*B*		

1/ When "*" is used for coded values, it's the contractors responsibility to select those options allowed by the CID.

2/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID P/N shown.

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7.6 Government users. To acquire information on obtaining these coils from the Government inventory system, contact Defense Supply Center, Columbus, ATTN: DSCC-CPAA, Post Office Box 3990, Columbus, OH 43216-5000, or telephone (614) 692-7741.

MILITARY INTERESTS:

Custodians:
Navy - EC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 7FXE

Preparing Activity:

DLA - CC

Project 5950-1162

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